

**Quantifying SZZ-Induced Label Noise in Just-In-Time Software
Defect Prediction under Verification-Latency-Aware Online
Evaluation**

A

report submitted in fulfilment for the award of the degree of

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In

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DECLARATION

I hereby certify that the work, which is being presented in the report/thesis, entitled Quantifying SZZ-Induced Label Noise in Just-In-Time Software Defect Prediction under Verification-Latency-Aware Online Evaluation in fulfilment of the requirements for the award of the degree of Master of Technology and submitted to the institution is an authentic record of my own work carried out during the period July 2026 to October 2026 under the supervision of Dr. Santosh Singh Rathore.

I also cited the reference about the text(s)/figure(s)/table(s) found where they have been taken.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dated:

Signature of the Supervisor

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Abstract

Just-in-time software defect prediction (JIT-SDP) aims to identify defect-introducing commits at the time they are submitted, so that review and testing effort can be directed to the changes that are most likely to be faulty. The models used for this task are trained on historical commits whose labels are almost always produced by the SZZ algorithm. SZZ starts from a bug-fixing commit and traces the modified lines back through the version history to find the commits that introduced the defect. This procedure is known to make mistakes, mainly because bug-fixing commits often contain changes that are unrelated to the fix. Since the same SZZ labels are generally used for training and for testing, the effect of these mistakes is not visible in the usual experimental setup.

The aim of this thesis is to measure how the label noise introduced by SZZ affects JIT-SDP models when they are evaluated under realistic conditions, that is, when the order of the commits and the delay with which labels become available are respected. This report presents the work carried out in the first stage of the thesis. The literature on JIT-SDP, on the SZZ algorithm and its variants, on label noise in defect prediction, on the evaluation of prediction models and on online learning with delayed labels is reviewed. From this review, five research gaps are identified, and four research questions and six objectives are formulated. A dataset that suits the study is then selected and described. It is the JIT-Defects4J dataset, which contains 27,319 commits from 21 Apache Java projects together with reference labels that were derived from manually validated bug-fixing lines. With such labels, the SZZ labels and the predictions of the models can be assessed against a source that does not depend on the SZZ variants under study.

The report ends with the plan of work for the remaining stages. In the second stage the labels of six SZZ variants will be compared with the reference labels, and their effect on prediction models will be measured under several evaluation settings. The third and fourth stages will study how the noise affects an online learner and how its effect can be reduced.

Keywords: just-in-time software defect prediction, SZZ algorithm, label noise, verification latency, online learning, class imbalance

Contents

Declaration	i
Acknowledgments	ii
Abstract	iii
List of Abbreviations	viii
1 Introduction	1
1.1 Background	1
1.2 Labelling of Commits and the SZZ Algorithm	2
1.3 Motivation	3
1.4 Problem Statement	4
2 Literature Review	6
2.1 Software Defect Prediction	6
2.2 Just-in-Time Software Defect Prediction	7
2.2.1 Change-level prediction and change metrics	7
2.2.2 Learning techniques	7
2.3 Identification of Defect-Introducing Changes	8
2.3.1 The SZZ algorithm	8
2.3.2 Variants of SZZ	8
2.3.3 Empirical evaluations of SZZ	9
2.3.4 Tangled changes	10
2.4 Label Noise in Defect Prediction	10
2.5 Evaluation of Defect Prediction Models	11
2.5.1 Validation schemes and time order	11
2.5.2 Prequential evaluation	11
2.5.3 Verification latency	12
2.5.4 Performance measures	12
2.6 Online Learning for Imbalanced Data Streams	13
2.7 Learning with Noisy Labels	13
2.8 Summary	14
3 Research Gaps, Research Questions and Objectives	15

3.1	Research Gaps	15
3.2	Research Questions	16
3.3	Objectives	17
4	Dataset	18
4.1	Requirements	18
4.2	The JIT-Defects4J Dataset	19
4.3	Subject Projects	19
4.4	Reference Labels	19
4.5	Change Metrics	21
4.6	Data Preparation	22
4.7	Information to be Added to the Dataset	23
4.8	Summary	23
5	Conclusion and Plan of Work	24
5.1	Summary of the Work Done	24
5.2	Plan of Work	25
5.2.1	Second stage: label quality and impact on prediction models . . .	25
5.2.2	Third stage: mechanism of the noise effect	26
5.2.3	Fourth stage: noise-aware online learner	26
5.3	Timeline	26
	References	27

List of Figures

1.1	General workflow of just-in-time software defect prediction.	2
1.2	Example of how SZZ identifies defect-introducing commits and how a change that is unrelated to the fix leads to a wrong label.	3
4.1	Number of commits and percentage of defect-introducing commits in each project. The dashed line marks the percentage in the whole dataset (8.54%).	21
5.1	Planned studies of the thesis and the research questions they address. . . .	25

List of Tables

2.1	SZZ variants considered in this work.	9
2.2	Summary of the most closely related studies.	14
4.1	Projects of the JIT-Defects4J dataset. The last two columns give the number and the percentage of defect-introducing commits according to the reference labels.	20
4.2	The fourteen change metrics used as features [3].	22
5.1	Timeline of the thesis work.	26

List of Abbreviations

AG-SZZ	Annotation Graph SZZ
B-SZZ	Basic SZZ (the original algorithm)
JIT-SDP	Just-In-Time Software Defect Prediction
L-SZZ	Largest SZZ
MA-SZZ	Meta-change Aware SZZ
MCC	Matthews Correlation Coefficient
OOB	Oversampling Online Bagging
ORB	Oversampling Rate Boosting
R-SZZ	Recent SZZ
RA-SZZ	Refactoring Aware SZZ
SZZ	Śliwerski–Zimmermann–Zeller algorithm

Chapter 1

Introduction

This chapter gives the background of the work. It introduces just-in-time software defect prediction, explains how the training labels for this task are obtained, and describes the issues that motivated the present study. The problem statement is given at the end of the chapter.

1.1 Background

Software systems are changed continuously during their lifetime, and some of these changes introduce defects. A defect that is found during code review can usually be corrected quickly by the developer who wrote the code. If the same defect is found after release, it has to be reported, reproduced, assigned and fixed, and it may already have affected the users of the system. For this reason, software quality assurance activities such as code review and testing try to find defects as early as possible. These activities need time and people, and in most projects it is not possible to review and test every part of the code with the same care.

Software defect prediction was proposed to help in this situation. A defect prediction model learns from the history of a project which parts of the code are likely to be defective, so that the available effort can be spent on those parts first [1, 2]. Most of the early work predicts defects at the level of files, classes or modules, and the prediction is generally made for a release of the software. Such models are useful for planning, but they have some practical limitations. A file that is predicted as defect-prone may be large, so the developer still has to find the faulty part. The prediction is made late in the development cycle, when the changes that caused the problem are no longer fresh in the mind of the developer. It is also not always clear who should inspect a file that has been modified by many developers [3].

Just-in-time software defect prediction (JIT-SDP) addresses these limitations by making the prediction for each code change (commit) instead of each file. When a developer

submits a commit, the model estimates whether this commit is likely to introduce a defect. The prediction is available immediately, it concerns a small amount of code, and it can be assigned to the author of the commit. The idea goes back to the work of Mockus and Weiss [4] and Kim et al. [5], and the formulation that is commonly used today was given by Kamei et al. [3]. Figure 1.1 shows the general workflow. The past commits of a project are described by a set of change metrics and are labelled as clean or defect-introducing. A classifier is trained on these commits and is then applied to every new commit.

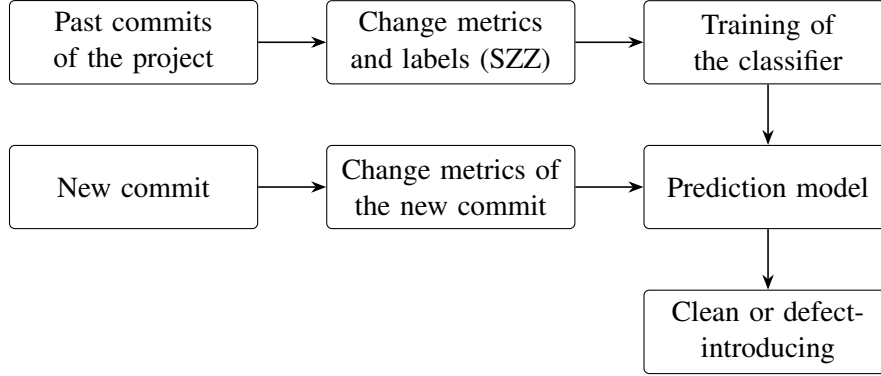


Figure 1.1: General workflow of just-in-time software defect prediction.

Three characteristics of this task are important for the present work. First, commits arrive one after another in time, so the data form a stream and the model is always used on commits that are newer than the ones it was trained on. Second, only a small fraction of the commits introduce defects. In the dataset used in this work the fraction is 8.54%, which means that the data are highly imbalanced. Third, the label of a commit is not known when the commit is made. It becomes known only later, when the defect is noticed and fixed, and for a clean commit it is never confirmed at all.

1.2 Labelling of Commits and the SZZ Algorithm

To train a JIT-SDP model, every past commit needs a label that says whether it introduced a defect. Software repositories do not store this information. What can be found in a repository and its issue tracker is the set of commits that *fixed* defects, because developers normally mention the bug report in the commit message. The commits that *introduced* the defects have to be inferred from these bug-fixing commits.

The standard method for this inference is the SZZ algorithm, proposed by Śliwerski, Zimmermann and Zeller [6]. The algorithm takes a bug-fixing commit and looks at the lines that were deleted or modified by it. For each of these lines it uses the annotation facility of the version control system (`git blame`) to find the earlier commit in which the line was last changed. These earlier commits are labelled as defect-introducing. All commits that are never reached in this way are treated as clean.

Figure 1.2 illustrates the procedure with a small example. Commit c_2 adds a faulty line. The fault is corrected later in the bug-fixing commit c_5 . When SZZ traces the corrected line back, it reaches c_2 , which is the right answer. However, the developer of c_5 has also edited another line that had nothing to do with the bug, for example to improve the formatting. This line was last changed in c_3 , so SZZ labels c_3 as defect-introducing as well, although it is clean.

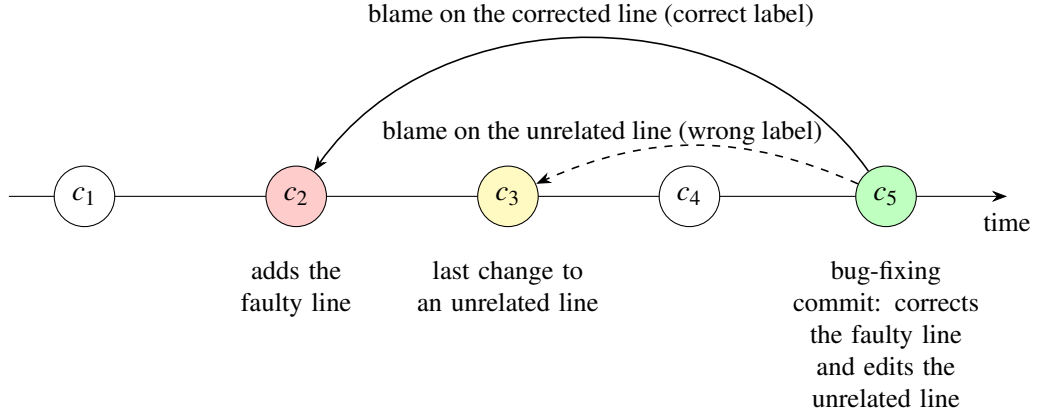


Figure 1.2: Example of how SZZ identifies defect-introducing commits and how a change that is unrelated to the fix leads to a wrong label.

A bug-fixing commit that contains such unrelated modifications is called a *tangled* commit. Tangled commits are common. Herbold et al. [7] manually examined a large number of bug-fixing commits and found that only 17% to 32% of the changed lines in them actually contribute to the bug fix. The remaining lines are changes to tests, documentation, formatting, refactoring and other modifications. Several improved versions of SZZ have been proposed to remove such lines before the blame step, and they are reviewed in Chapter 2. It is important to note that all of them still *estimate* the label. None of the labels used in JIT-SDP studies is directly observed.

1.3 Motivation

The motivation for this work comes from four issues in the way JIT-SDP models are currently built and evaluated.

Noise in the SZZ labels. Since the labels are produced by a heuristic, some clean commits are labelled as defect-introducing (false positives) and some defect-introducing commits are labelled as clean (false negatives). A model trained on such labels learns partly from wrong examples. Earlier studies have evaluated the accuracy of SZZ on small manually checked samples [8, 9], but the amount and the type of noise over the complete history of a project, which is what a JIT-SDP model is trained on, has received less attention.

Evaluation against the same noisy labels. In nearly all JIT-SDP studies the test set is labelled by the same SZZ implementation as the training set. A model is therefore rewarded when it predicts what SZZ would say, including the cases where SZZ is wrong. The reported performance then measures how well the model reproduces the labelling heuristic, which is not the same as how well it finds real defects. The difference between the two can only be measured if labels from an independent source are available for testing.

Evaluation that ignores time. Many studies evaluate their models with random k -fold cross-validation. On commit data this means that the model may be trained on commits that were made after the commits on which it is tested. Such a setting cannot occur in practice, and it has been shown for defect prediction in general that it gives optimistic results [10, 11].

Delay in the arrival of labels. Even when the commits are split by time, it is normally assumed that the label of every training commit is known at training time. In reality a defect-introducing commit is recognised only when its defect is fixed, which may happen months or years later. Cabral et al. [12] called this delay *verification latency*. During the delay the commit is treated as clean, so a model that is updated continuously receives a wrong label first and the correct label later. The delay can be long. Cabral and Minku [13] report that the time needed to discover a defect ranges from one day to more than eleven years.

These four issues are related. The first one is about the quality of the training data, and the other three are about the realism of the evaluation. A study that wants to know how much the label noise matters in practice has to control all of them at the same time.

1.4 Problem Statement

JIT-SDP models are trained and evaluated on labels generated by the SZZ algorithm, and the evaluation is often carried out without respecting the order of commits and the delay with which labels become available. As a result, it is not known how much the errors made by SZZ reduce the ability of a model to identify real defect-introducing commits when the model is used in a realistic setting, and it is not possible to say how much of the performance reported in the literature is due to the model and how much is due to the way it was evaluated.

The problem addressed in this thesis is therefore to quantify the label noise introduced by SZZ and its variants, to measure the effect of this noise on JIT-SDP models under an evaluation that respects time order and verification latency, and, in the later part of the

thesis, to understand how the noise affects an online learner and how its effect can be reduced.

The thesis work is divided into four stages. This report presents the work of the first stage. It consists of the literature survey, the identification of the research gaps, the formulation of the research questions and objectives, and the selection and description of the dataset. The work planned for the remaining stages is described in Chapter 5.

Chapter 2

Literature Review

This chapter reviews the literature related to the present work. It starts with defect prediction in general and JIT-SDP in particular, then discusses the SZZ algorithm and its variants, the studies on label noise, the evaluation of prediction models, online learning for imbalanced data streams and learning with noisy labels. A summary of the most closely related studies is given at the end of the chapter.

2.1 Software Defect Prediction

Software defect prediction has been studied for more than three decades. The usual approach is to describe each software module by a set of metrics, such as size, complexity, coupling or the history of past changes, and to train a classifier that separates defective modules from clean ones. Hall et al. [1] carried out a systematic review of 208 fault prediction studies and observed that models based on simple techniques such as Naïve Bayes and logistic regression often perform as well as more complex ones, and that the quality of the data and of the experimental procedure has a large influence on the reported results. Rathore and Kumar [2] reviewed the field with respect to three aspects: the software metrics used, the prediction techniques and the data quality issues. Among the data quality issues they discuss class imbalance and noise in the data, both of which are central to the present work.

Most of this research predicts defects for files, classes or modules of a given release. The limitations of this granularity, mentioned in Chapter 1, led to the prediction of defects at the level of individual changes.

2.2 Just-in-Time Software Defect Prediction

2.2.1 Change-level prediction and change metrics

Mockus and Weiss [4] were among the first to predict the risk of a software change. They used properties of the change, such as its size, the number of files and subsystems it touches, the experience of the developer and the type of the change, to predict the probability that a change to a large telecommunication system would cause a failure. Kim et al. [5] introduced *change classification*, in which a classifier is trained to decide whether a change is buggy or clean. They used features extracted from the change log, the source code and the change metadata, and obtained on average 78% accuracy and 60% recall for buggy changes on twelve open-source projects.

Kamei et al. [3] carried out a large empirical study on six open-source and five commercial projects and introduced the term “just-in-time” for this kind of quality assurance. They proposed fourteen change metrics that are grouped into five dimensions: diffusion, size, purpose, history and experience. Their logistic regression model predicted defect-inducing changes with an average accuracy of 68% and a recall of 64%. These fourteen metrics have become the standard feature set for JIT-SDP and are also used in the present work. They are described in detail in Section 4.5. A recent survey by Zhao et al. [14] covers 67 JIT-SDP studies and shows that the majority of them follow the formulation of Kamei et al. with respect to both the features and the way the labels are obtained.

McIntosh and Kamei [15] studied how JIT-SDP models behave over time on the Qt and OpenStack projects. They found that the properties of defect-inducing changes are not stable. The importance of the different metrics changes as a project evolves, and a model loses a considerable part of its predictive power when it is used long after it was trained. This result shows that time has to be taken into account both in training and in evaluating such models.

2.2.2 Learning techniques

A wide range of learning techniques has been applied to JIT-SDP. Besides logistic regression, random forests and other ensemble methods are frequently used. More recently, deep learning models have been proposed that learn features directly from the content of a change. DeepJIT [16] applies convolutional networks to the commit message and to the changed code. CC2Vec [17] learns a vector representation of a code change with a hierarchical attention network and combines it with DeepJIT.

Later studies have questioned whether this additional complexity is necessary. Zeng et al. [18] replicated DeepJIT and CC2Vec on a larger dataset of more than 310,000 changes. They reported that the deep models did not consistently perform better than traditional ap-

proaches, and that a logistic regression model using only the number of added lines, which they called LAPredict, outperformed both of them in most cases while being many times faster. Pornprasit and Tantithamthavorn [19] proposed JITLine, which uses a random forest together with oversampling of the minority class, and showed that it is more accurate and much faster than the deep models. Yang et al. [20] had earlier shown for effort-aware prediction that simple unsupervised models built from a single change metric can be better than supervised models.

These studies are relevant here for two reasons. First, they show that the ranking of models in JIT-SDP depends strongly on how the evaluation is done. Second, they provide two well-known models of very different complexity, LAPredict and JITLine, which will be used in the present work to examine whether the effect of label noise depends on the capacity of the model.

2.3 Identification of Defect-Introducing Changes

2.3.1 The SZZ algorithm

The SZZ algorithm [6] consists of two phases. In the first phase, bug-fixing commits are identified by linking the commits of the version control system to the reports in the bug tracking system, mainly by searching the commit messages for bug identifiers. In the second phase, for every bug-fixing commit the lines that were deleted or modified are determined, and the `annotate (blame)` command is used to find the most recent earlier commits that changed these lines. Commits that were made after the bug was reported are excluded, since they cannot have caused the bug. The remaining commits are considered to be bug-introducing.

The algorithm depends on two assumptions. The first is that the lines changed by the fix are the lines that contained the defect. The second is that the commit that last changed such a line is the commit that made it defective. Both assumptions can fail. The first one fails for tangled commits, where some of the changed lines are not related to the fix. The second one fails when a line was last touched by a cosmetic change, such as a renaming or re-indentation, so that the blame points to this change instead of the original faulty one. In addition, a defect that is fixed only by adding new lines leaves no deleted or modified line to trace, and a defect caused by a change in the environment or in the requirements has no introducing commit in the repository at all [21].

2.3.2 Variants of SZZ

Several variants have been proposed to reduce the number of wrong labels produced by the original algorithm, which is usually called B-SZZ (basic SZZ). Table 2.1 lists the

variants that are studied in this work.

Table 2.1: SZZ variants considered in this work.

Variant	Main idea
B-SZZ [6]	Original algorithm. Every line deleted or modified by the bug-fixing commit is traced back with blame.
AG-SZZ [22]	Uses an annotation graph to follow lines across revisions and ignores blank lines, comments and formatting changes.
MA-SZZ [8]	Builds on AG-SZZ and additionally ignores meta-changes, i.e. branch changes, merge changes and property changes, which do not modify the source code.
R-SZZ [23]	Builds on MA-SZZ and keeps, for each bug-fixing commit, only the most recent of the candidate commits.
L-SZZ [23]	Builds on MA-SZZ and keeps only the candidate commit that changed the largest number of lines.
RA-SZZ [24]	Builds on MA-SZZ and ignores lines that were changed by refactoring operations, which are detected with a refactoring detection tool. It is available for Java only.

Kim et al. [22] observed that many lines changed in a bug-fixing commit are comments, blank lines or formatting changes and proposed to ignore them. They also replaced the simple `annotate` command by an annotation graph, which follows a line more reliably when code is moved. Da Costa et al. [8] noticed that some commits flagged by SZZ are only changes to the repository structure, such as merges, and excluded these meta-changes. R-SZZ and L-SZZ go one step further and select a single commit from the candidates returned for a bug-fixing commit [23]. They assume that a bug has one introducing commit and choose either the latest or the largest candidate. Neto et al. [24] showed that refactoring operations are a frequent cause of wrong labels and proposed RA-SZZ, which removes refactored lines from the analysis.

All these variants try to reduce false positives by removing lines or candidate commits. An obvious consequence, which will be examined in this thesis, is that some correct candidates may be removed as well.

2.3.3 Empirical evaluations of SZZ

The accuracy of SZZ is difficult to evaluate, because the true defect-introducing commits are not known. Da Costa et al. [8] proposed a framework that evaluates SZZ implementations indirectly through three criteria: the earliest appearance of the bug, the future impact of the flagged changes and the realism of the bug introduction. They applied it to five implementations on ten open-source projects and found that the existing implementations still have considerable room for improvement.

Rodríguez-Pérez et al. [25] reviewed 187 papers that use SZZ and found that most

of them neither discuss the limitations of the algorithm nor provide enough detail to reproduce the labelling. In a later study [21] they manually analysed how 116 bugs were introduced in two open-source projects. They found that a part of the bugs are *extrinsic*, i.e. caused by changes outside the project code, and that the SZZ-based algorithms reached F-scores of only 0.44 to 0.77 on the remaining ones.

Rosa et al. [9, 26] built a *developer-informed oracle* from commit messages in which the developer explicitly names the commit that introduced the bug being fixed. The oracle contains 1,930 such cases, extended to 2,304 in the journal version, and was used to compare several SZZ variants. R-SZZ obtained the best results, with a precision of about 66% and an F-measure of about 61%, while B-SZZ had a higher recall but a precision of only about 39%. The authors also released PySZZ, an open-source implementation of the variants, which will be used in the present work.

Two properties of this oracle are important for the present study. It contains only bug-fixing commits together with their introducing commits and no clean commits, so it cannot be used to train or test a JIT-SDP model. It also evaluates SZZ only on fixes for which an introducing commit is known to exist. This is a different question from the one that matters for JIT-SDP, where every commit in the history of the project receives a label.

2.3.4 Tangled changes

Herzig and Zeller [27] studied tangled changes in five open-source Java projects and found that up to 15% of the bug fixes consist of several unrelated changes. As a result, at least 16.6% of the source files were incorrectly associated with bug reports. Herbold et al. [7] conducted a large crowd-sourced study in which every changed line of the bug-fixing commits of several Apache projects was labelled by four participants. They found that only 17% to 32% of all changed lines contribute to the bug fix. When only changes to production code files are considered, the share is 66% to 87%. For about 11% of the lines the participants could not agree on a label. The dataset produced by this study, called LLTC4J, is the basis of the reference labels used in the present work (Section 4.4).

2.4 Label Noise in Defect Prediction

The effect of wrong labels on defect prediction models has been studied mainly for file-level prediction. Kim et al. [28] added artificial false positive and false negative noise to defect datasets and found that the prediction performance decreases significantly once the datasets contain about 20% to 35% of both types of noise. They also proposed a method to detect and remove noisy instances. Tantithamthavorn et al. [29] examined mislabelling that arises from wrongly classified issue reports. They showed that such mislabelling is

not random, and that models trained on the noisy data keep their precision but reach only 56% to 68% of the recall of models trained on clean data.

For JIT-SDP, the most closely related study is that of Fan et al. [30]. They labelled 126,526 changes from ten Apache projects with four SZZ variants (B-SZZ, AG-SZZ, MA-SZZ and RA-SZZ) and examined how the differences between the variants affect JIT-SDP models. They reported that the changes mislabelled by B-SZZ and MA-SZZ do not reduce the prediction performance considerably, whereas those mislabelled by AG-SZZ lead to a statistically significant reduction. Since no ground truth was available, they treated the labels of RA-SZZ as correct and measured the mislabelling of the other variants relative to it. This has two consequences. The noise of RA-SZZ itself could not be measured, and errors that are common to all variants, for example those caused by tangled changes, remained invisible. The study also did not consider the delayed arrival of labels.

2.5 Evaluation of Defect Prediction Models

2.5.1 Validation schemes and time order

Cross-validation with random folds is the most common validation scheme in defect prediction. For data that are ordered in time it has a known weakness, because a model can be trained on data that are newer than the test data. Tan et al. [11] pointed out for change classification that cross-validation uses information from the future and can therefore report a precision that is too high. They proposed a time-sensitive evaluation in which the model is trained only on changes that precede the test changes. Falessi et al. [10] analysed this issue systematically for within-project defect classifiers. They showed that validation techniques which do not preserve the order of the data produce results that differ from those of order-preserving techniques, and recommended the latter. In spite of these findings, random cross-validation is still widely used in JIT-SDP studies [14].

2.5.2 Prequential evaluation

When data arrive as a stream, the standard evaluation procedure is *prequential* (predictive sequential) evaluation, also known as test-then-train. Each example is first used to test the model and afterwards to update it. The performance is accumulated over the stream. Since the performance at the beginning of the stream is often poor and less relevant, a forgetting mechanism such as a sliding window or a fading factor is applied. Gama et al. [31] showed that the prequential error computed with a fading factor converges to the same value as the error on an independent holdout set. This result holds for the error rate. For measures that are computed from the complete confusion matrix, such as MCC, no

corresponding result is available, which has to be kept in mind when prequential values of such measures are summarised.

2.5.3 Verification latency

Cabral et al. [12] were the first to investigate JIT-SDP as an online learning problem with verification latency. In their setting a commit is used as a clean training example when a waiting period has passed without a defect being linked to it, and as a defect-inducing example as soon as a defect is linked to it. They used a waiting period of 90 days. They showed that, because of this mechanism, the proportion of the two classes seen by the learner changes over time, which they called class imbalance evolution, and that this hinders the existing approaches. To deal with it they proposed the ORB method, which is described in Section 2.6. In a later study, Cabral and Minku [13] examined concept drift in JIT-SDP in more detail. They reported that the time needed to discover a defect ranges from one day to more than eleven years, and proposed a method that also uses the predictions on unlabelled commits to detect when the model has become unreliable. Tabassum et al. [32] studied the use of data from other projects in the same online setting.

In these studies the wrong labels that the learner receives are caused by time: a defect-introducing commit is treated as clean until its defect is found. The labels that finally arrive are assumed to be correct. The noise that comes from the SZZ algorithm itself, where the final label is wrong and is never corrected, is not addressed.

2.5.4 Performance measures

Because the classes are imbalanced, accuracy is not a suitable measure for defect prediction. Precision, recall and the F-measure are widely used, but the F-measure ignores the true negatives and depends on the proportion of the positive class. Yao and Shepherd [33] showed that the use of the F-measure can lead to different conclusions than the Matthews correlation coefficient (MCC) [34] in a considerable number of defect prediction comparisons and recommended MCC. Chicco and Jurman [35] reached the same recommendation for binary classification in general. MCC uses all four cells of the confusion matrix and takes the value zero for a classifier that predicts at random or always predicts the same class. For measuring the agreement between two sets of labels, Cohen's κ [36] is commonly used, which corrects the observed agreement for the agreement expected by chance. Landis and Koch [37] proposed the widely used verbal interpretation of κ values, according to which values up to 0.20 indicate slight agreement and values above 0.80 almost perfect agreement.

2.6 Online Learning for Imbalanced Data Streams

Online Bagging [38] is the online version of the bagging ensemble. In ordinary bagging each base learner is trained on a bootstrap sample of the data. In the online version, each arriving example is presented to each base learner k times, where k is drawn from a Poisson distribution with parameter $\lambda = 1$. Wang et al. [39] adapted this method to imbalanced streams. In their Oversampling Online Bagging (OOB), the parameter λ for an example of the minority class is increased according to the current ratio of the class sizes, so that minority examples are presented more often. The class sizes are tracked with a time-decayed average, which allows the method to follow changes in the imbalance.

ORB (Oversampling Rate Boosting) [12] extends OOB for JIT-SDP. It monitors the predictions that the model has made recently. If the model predicts one class much more often than expected, the resampling rate of the other class is increased further by a boosting factor. In this way the model is pushed back when it becomes biased towards one class, which happens frequently under verification latency.

It should be noted that in all these methods the resampling rate depends only on how many examples of each class have been observed. Whether the labels of these examples are correct plays no role. A commit that is wrongly labelled as defect-introducing is oversampled in the same way as a correctly labelled one. This property is one of the reasons why the effect of label noise on such learners deserves to be studied.

2.7 Learning with Noisy Labels

In the machine learning literature, several methods have been proposed for training classifiers on data with wrong labels. Natarajan et al. [40] considered the case where the probability that a label is flipped depends on the class, which is called class-conditional noise. They showed that, if the two flip rates are known, the loss function can be modified such that minimising it on the noisy data is equivalent in expectation to minimising the original loss on clean data. The correction contains the factor $1/(1 - \rho_0 - \rho_1)$, where ρ_0 and ρ_1 are the two flip rates, and becomes unstable when the sum of the rates approaches one. Co-teaching [41] trains two neural networks at the same time. Each network selects the examples with small loss in a mini-batch, which are likely to be correctly labelled, and passes them to the other network for training. Confident learning [42] estimates the joint distribution of the given and the true labels from the predicted probabilities of a classifier and uses it to identify and remove the examples that are probably mislabelled.

These methods were developed for the batch setting, in which the complete dataset is available and can be processed several times. They also assume that a wrong label stays wrong. In online JIT-SDP each example is seen once, its label arrives late, and a label

may be changed later from clean to defect-introducing. The methods can therefore not be applied directly, but the class-conditional noise model on which they are based is the one that will be measured in the present work.

2.8 Summary

Table 2.2 summarises the studies that are most closely related to the present work and indicates in which respect each of them differs from it. The review shows that the accuracy of SZZ, the effect of label noise, time-aware evaluation and verification latency have each been studied, but mostly in isolation. The gaps that follow from this are stated in the next chapter.

Table 2.2: Summary of the most closely related studies.

Study	Focus	Difference from the present work
Kamei et al. [3]	Change metrics and models for JIT-SDP	Labels from SZZ are taken as correct; no time-aware evaluation
Da Costa et al. [8]	Framework for evaluating SZZ implementations	Indirect criteria; effect on prediction models not studied
Rosa et al. [9, 26]	Accuracy of SZZ variants against a developer-informed oracle	Oracle has no clean commits; effect on prediction models not studied
Herbold et al. [7]	Amount of tangling in bug-fixing commits	Line-level analysis; labels of commits and models not studied
Kim et al. [28]	Artificial label noise in file-level defect data	Noise is injected artificially; file-level prediction with batch evaluation
Fan et al. [30]	Effect of labels of four SZZ variants on JIT-SDP	RA-SZZ used as reference; no independent labels; no latency
Falessi et al. [10]	Order-preserving validation of defect classifiers	Label noise not considered
Cabral et al. [12]	Online JIT-SDP with verification latency and class imbalance evolution	SZZ labels taken as correct once they arrive
Zeng et al. [18]	Comparison of deep and simple JIT-SDP models	Label noise and latency not considered
Natarajan et al. [40]	Learning under class-conditional label noise	Batch setting; noise rates assumed to be known

Chapter 3

Research Gaps, Research Questions and Objectives

This chapter states the research gaps that were identified from the literature reviewed in Chapter 2. Based on these gaps, the research questions and the objectives of the thesis are formulated. The stage of the thesis in which each of them will be addressed is indicated.

3.1 Research Gaps

The following gaps were identified.

Gap 1: The noise of SZZ labels has not been measured over complete project histories. The existing evaluations of SZZ either use indirect criteria [8] or check the algorithm on a sample of bug-fixing commits for which the introducing commit is known [9, 21]. These evaluations tell how often SZZ finds the right commit for a given fix. They do not tell how many of the commits in the history of a project receive a wrong label, which is the quantity that matters for a JIT-SDP model, because the model is trained on all commits. In particular, the rate of false positives among the clean commits and the rate of false negatives among the defect-introducing commits have not been reported separately for the different variants.

Gap 2: Models are tested against the same labels on which they are trained. In the JIT-SDP literature the test labels come from the same SZZ implementation as the training labels. The study that is closest to the present work [30] used one SZZ variant as the reference for the others. Hence it is not known how well a model that was trained on SZZ labels predicts labels that were obtained independently of the SZZ variant used for training.

Gap 3: The effect of label noise has not been separated from the effect of the evaluation procedure. It is known that random cross-validation gives optimistic results for time-ordered data [10, 11], and it is known that SZZ labels contain errors. These two factors are present together in most published results, and their individual contributions have not been measured in a common experimental design.

Gap 4: Label noise has not been studied under verification latency. The studies on online JIT-SDP [12, 13] model the delay of the labels but assume that the labels which finally arrive are correct. The studies on label noise use batch evaluation [28, 30]. The loss caused by SZZ noise for an online learner that receives delayed labels is therefore not known.

Gap 5: The way in which SZZ noise affects an online learner is not understood, and no remedy designed for this setting exists. Online learners for imbalanced streams oversample the minority class according to the number of positive labels they observe (Section 2.6), without regard to their correctness. Whether false positives or false negatives are more harmful for such learners is not known. The existing methods for learning with noisy labels assume a batch setting and stable labels [40, 42] and cannot be applied directly to a stream with delayed labels.

3.2 Research Questions

Based on these gaps, the thesis addresses the following research questions.

RQ1: How much do the labels of different SZZ variants disagree with one another and with independently obtained reference labels, and what type of error does each variant make? This question corresponds to Gap 1. To answer it, the labels of six SZZ variants will be compared with reference labels on the same set of commits, and the false positive and false negative rates of each variant will be estimated. This question will be addressed in the second stage of the thesis.

RQ2: How does the choice of the label source affect the measured performance of JIT-SDP models under different evaluation settings? This question corresponds to Gaps 2, 3 and 4. To answer it, models will be trained on the reference labels and on the labels of each SZZ variant, and will be evaluated with random cross-validation, with a chronological split and with prequential evaluation under verification latency. The predictions will be scored against the reference labels and, for comparison, against the training labels. This question will also be addressed in the second stage.

RQ3: Through which mechanism does label noise reduce the performance of an online learner, and which of the two types of error is responsible? This question corresponds to the first part of Gap 5. It will be addressed in the third stage.

RQ4: Can an online learner be modified so that it loses less performance when it is trained on SZZ labels, without access to clean labels and without losing performance when the labels are clean? This question corresponds to the second part of Gap 5. It will be addressed in the fourth stage.

3.3 Objectives

The objectives of the thesis are as follows.

1. To generate the labels of six SZZ variants for all projects of a dataset that provides independent reference labels, and to characterise the labelling errors of each variant in terms of false positive and false negative rates.
2. To examine the agreement between the SZZ variants and to identify the sources of the errors that the variants make.
3. To measure how much the evaluation procedure, namely the validation scheme and the labels used for scoring, changes the measured performance of JIT-SDP models.
4. To measure the loss in performance caused by training on SZZ labels when an online learner is evaluated on a commit stream with verification latency.
5. To identify the type of labelling error that is responsible for this loss and the mechanism through which it acts.
6. To design and evaluate a modification of the online learner that reduces this loss.

Objectives 1 to 4 will be addressed in the second stage, objective 5 in the third stage and objective 6 in the fourth stage. The plan of work for these stages is given in Section 5.2.

Chapter 4

Dataset

This chapter describes the dataset used in the study. It first states the requirements that the dataset has to satisfy and then describes the JIT-Defects4J dataset, the projects it contains, the reference labels, the change metrics and the preparation of the data.

4.1 Requirements

The research questions place several requirements on the dataset.

- A label must be available for every commit of a project, including the clean commits, because JIT-SDP models are trained and tested on the complete commit history.
- The labels must not have been produced by the SZZ variant that is being evaluated. Otherwise the errors of that variant cannot be observed.
- The change metrics of the commits must be available, so that prediction models can be trained.
- The repositories of the projects must be accessible, because the SZZ variants have to be executed on them and the dates of the bug-fixing commits are needed for simulating verification latency.
- The dataset should contain several projects, so that the comparisons can be tested statistically across projects.

Most JIT-SDP datasets do not satisfy the second requirement. ApacheJIT [43], which contains 106,674 commits, and Defectors [44] are much larger than the dataset used here, but their labels were generated with SZZ. The developer-informed oracle of Rosa et al. [9] does not satisfy the first requirement, since it contains no clean commits. The JIT-Defects4J dataset satisfies all five requirements and was therefore selected.

4.2 The JIT-Defects4J Dataset

JIT-Defects4J was published by Ni et al. [45] together with their JIT-Fine approach. It was built from LLTC4J, the dataset of Herbold et al. [7], in which the changed lines of bug-fixing commits were labelled manually. JIT-Defects4J contains 27,319 commits from 21 open-source Java projects of the Apache Software Foundation. For every commit it provides the fourteen change metrics of Kamei et al. and a label that indicates whether the commit introduced a defect. The dataset was used as it was published. No project was added or removed, so the selection of projects could not be influenced by the results of this study.

4.3 Subject Projects

Table 4.1 lists the 21 projects. Sixteen of them are libraries of the Apache Commons family. The other five are a dependency manager (ant-ivy), a graph processing system (giraph), a data persistence framework (gora), a natural language processing toolkit (opennlp) and a columnar storage format (parquet-mr). The commits were made between 2001 and 2019. The length of the covered history of a project varies from about 6 years (parquet-mr) to about 18 years (commons-dbc).

In total, 5,453 commits (20.0%) are marked as bug-fixing and 2,332 commits (8.54%) are defect-introducing according to the reference labels. The projects differ considerably in size and in the proportion of defect-introducing commits, as can be seen in Figure 4.1. The smallest project has 544 commits and the largest has 4,026. The number of defect-introducing commits ranges from 19 in commons-digester to 335 in commons-math, and the defect rate ranges from 1.8% to 19.3%. In every project the defect-introducing commits are a clear minority. Because of these differences, all comparisons in this work will be made project by project, and the results will then be summarised over the projects.

4.4 Reference Labels

The label provided with JIT-Defects4J will be used in this work as the reference against which the SZZ variants and the predictions of the models are assessed. Since all later results will depend on this label, its construction is described in detail.

According to the description given by the authors of the dataset [45], the labels were obtained in two steps. In the first step, they took the bug-fixing commits of LLTC4J. In LLTC4J every changed line of a bug-fixing commit was labelled by four participants, and a line is regarded as contributing to the bug fix only if at least three of them assigned this label. In the second step, `git blame` was applied to each of these validated lines to find

Table 4.1: Projects of the JIT-Defects4J dataset. The last two columns give the number and the percentage of defect-introducing commits according to the reference labels.

Project	Period	Commits	Bug-fixing commits	Defect-introducing	
				Number	%
ant-ivy	2005–2018	1,771	761	332	18.7
commons-bcel	2001–2019	825	134	60	7.3
commons-beanutils	2001–2018	611	197	37	6.1
commons-codec	2003–2018	761	122	36	4.7
commons-collections	2001–2018	1,823	360	50	2.7
commons-compress	2003–2018	1,630	175	178	10.9
commons-configuration	2003–2018	1,838	358	155	8.4
commons-dbcp	2001–2019	1,037	348	58	5.6
commons-digester	2001–2017	1,079	221	19	1.8
commons-io	2002–2018	1,142	224	73	6.4
commons-jcs	2002–2018	831	237	88	10.6
commons-lang	2002–2018	2,969	491	146	4.9
commons-math	2003–2018	4,026	758	335	8.3
commons-net	2002–2018	1,121	197	117	10.4
commons-scxml	2005–2018	544	55	47	8.6
commons-validator	2002–2018	598	120	36	6.0
commons-vfs	2002–2018	1,110	201	114	10.3
giraph	2010–2018	844	118	163	19.3
gora	2010–2019	553	85	39	7.1
opennlp	2010–2018	1,086	83	91	8.4
parquet-mr	2012–2018	1,120	208	158	14.1
Total		27,319	5,453	2,332	8.54

the commit that introduced it. The commits found in this way are the defect-introducing commits. All other commits are considered clean.

To make sure that the labels in the files obtained for this work are indeed these labels, the number of commits and of defect-introducing commits of every project was compared with the numbers reported in the paper. They agree for all 21 projects.

The reference labels are thus the result of the blame operation applied to manually validated bug-fixing lines. They are not a manual judgement about each commit. This has the following consequences for the interpretation of the results.

1. The reference labels and the SZZ labels differ in the lines from which the tracing starts. SZZ starts from all lines changed in a bug-fixing commit (after the filtering applied by the particular variant), whereas the reference starts only from the lines that were confirmed to be part of the fix. A comparison between the two therefore shows the effect of tangled changes, and of the filters that the variants use to deal with them.

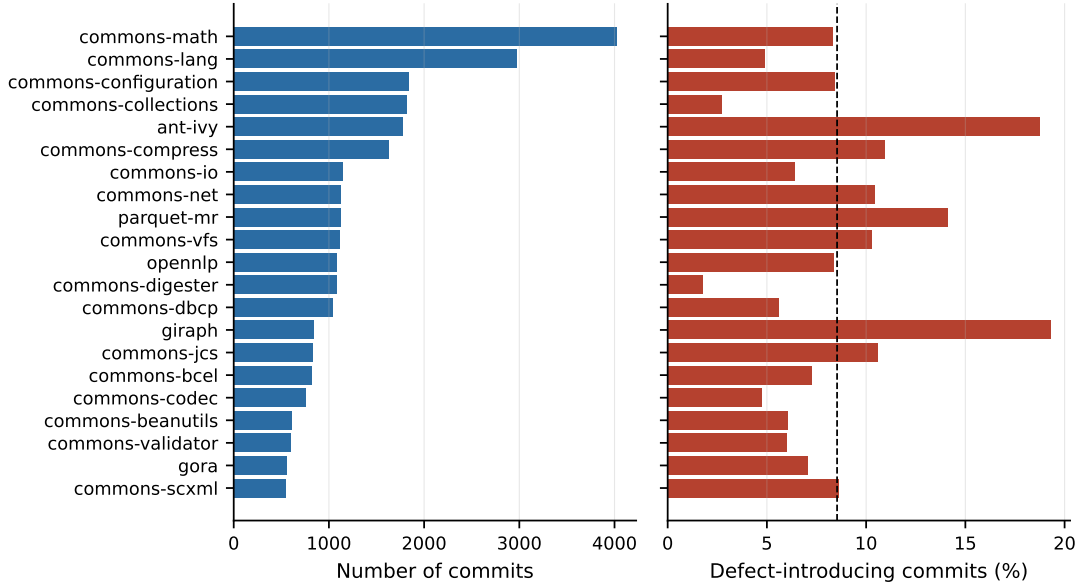


Figure 4.1: Number of commits and percentage of defect-introducing commits in each project. The dashed line marks the percentage in the whole dataset (8.54%).

2. Both use the blame operation to go from a line to a commit. Errors of the blame operation itself, for example when a line was last touched by a cosmetic change, are present in both and cannot be detected by comparing them. The amount of noise that will be measured in this work should therefore be regarded as a lower bound of the true amount.
3. Every commit that is defect-introducing according to the reference can be reached by blame from some line of a bug-fixing commit. When an SZZ variant misses such a commit, the reason cannot be that blame is unable to reach it. The reason must lie in the lines selected or in the filtering of the candidates.
4. The authors of the dataset excluded changes that do not add any line. Defects that are introduced only by deleting code are therefore not represented.

In the rest of this report the term *reference labels* is used for these labels. The term “ground truth” is avoided, because the labels are more reliable than SZZ labels with respect to tangling but are not free of error.

4.5 Change Metrics

Each commit is described by the fourteen change metrics proposed by Kamei et al. [3]. They are listed in Table 4.2. The metrics of the diffusion group measure how widely a change is spread over the code base, since changes that touch many parts are more difficult to understand. The size group measures how much code was changed. The purpose

group contains a single metric which indicates whether the change is a bug fix, since bug fixes are known to introduce new defects more often than other changes. The history group describes how the modified files were changed in the past, and the experience group describes how familiar the developer is with the project and with the modified subsystem.

Table 4.2: The fourteen change metrics used as features [3].

Group	Metric	Description
Diffusion	NS	Number of modified subsystems
	ND	Number of modified directories
	NF	Number of modified files
	Entropy	Distribution of the modified code across the files
Size	LA	Lines of code added
	LD	Lines of code deleted
	LT	Lines of code in the files before the change
Purpose	FIX	Whether the change is a defect fix
History	NDEV	Number of developers that changed the modified files
	AGE	Average time interval between the last and the current change
	NUC	Number of unique changes to the modified files
Experience	EXP	Number of changes made by the developer so far
	REXP	Recent experience of the developer (changes weighted by their age)
	SEXP	Experience of the developer on the modified subsystem

The values of the metrics were taken from the dataset as published and were not re-computed. In this way the features are identical for all label sources, and differences between the results obtained with different labels cannot be caused by differences in feature extraction. Several of the metrics have strongly skewed distributions. Suitable transformations will be applied to them when the prediction models are trained.

4.6 Data Preparation

The dataset is distributed in three files that correspond to the training, validation and test split used by its authors. These files were merged, since the present work uses its own evaluation settings. Duplicate records were removed on the combination of project and commit identifier. Within each project the commits were sorted by their author timestamp. This order defines the commit stream of the project and is used in all evaluation settings that depend on time.

The repositories of the 21 projects were also cloned from GitHub. They will be needed in the second stage to run the SZZ variants and to obtain the dates of the bug-fixing commits.

4.7 Information to be Added to the Dataset

The dataset provides one label for each commit, the reference label. For the purpose of this thesis two kinds of information have to be added to it. Both will be produced in the second stage of the thesis.

The first is the set of SZZ labels. To study the noise of SZZ, six further labels will be attached to every commit, one for each SZZ variant of Table 2.1. They will be generated by running the variants on the repositories of the 21 projects, with the bug-fixing commits of the dataset as input. Each commit will then have seven binary labels.

The second is the time at which the label of each commit becomes available. To evaluate an online learner realistically, it must be known when a commit could first have been recognised as defect-introducing. This is the time at which its defect was fixed. The dataset does not contain this information. It will be reconstructed from the dates of the bug-fixing commits, which can be read from the repositories.

4.8 Summary

The study will use the JIT-Defects4J dataset with 27,319 commits from 21 Apache Java projects, of which 8.54% are defect-introducing according to reference labels that were derived from manually validated bug-fixing lines. Each commit is described by fourteen change metrics. The files of the dataset were merged, cleaned and ordered by time, and the repositories of the projects were cloned. The labels of the SZZ variants and the arrival times of the labels will be added in the second stage.

Chapter 5

Conclusion and Plan of Work

This chapter summarises the work done in the first stage and describes the work planned for the remaining stages of the thesis.

5.1 Summary of the Work Done

The thesis deals with the effect of the label noise introduced by the SZZ algorithm on just-in-time software defect prediction. The work of the first stage consisted of three parts.

First, the literature was reviewed. The review covered defect prediction at the level of changes, the SZZ algorithm and its variants together with the studies that evaluate them, the studies on label noise in defect data, the evaluation of prediction models with respect to time and to the delayed arrival of labels, online learning for imbalanced data streams, and learning with noisy labels.

Second, the research gaps were identified and the research questions and objectives were formulated. The review showed that the accuracy of SZZ, the effect of label noise, time-aware evaluation and verification latency have each been studied, but mostly separately. The noise of SZZ labels has not been measured over complete project histories, models are tested against the same labels on which they are trained, the effect of label noise has not been separated from the effect of the evaluation procedure, and label noise has not been studied under verification latency. Four research questions and six objectives were derived from these gaps (Chapter 3).

Third, a dataset was selected and prepared. The JIT-Defects4J dataset was chosen because it provides, for every commit of 21 projects, a label that was derived from manually validated bug-fixing lines and not from the SZZ variants under study. The construction of these labels was examined, and its consequences for the planned comparisons were noted. The dataset was merged, cleaned and ordered by time, and the repositories of the projects were obtained.

5.2 Plan of Work

Figure 5.1 shows the studies that are planned and the research question that each of them addresses.

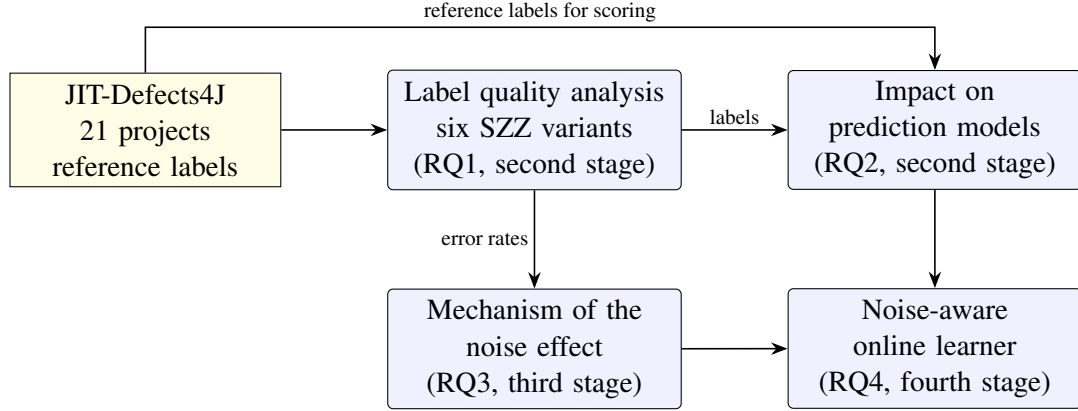


Figure 5.1: Planned studies of the thesis and the research questions they address.

5.2.1 Second stage: label quality and impact on prediction models

The work of the second stage addresses RQ1 and RQ2 and consists of two studies.

In the first study, the six SZZ variants of Table 2.1 will be executed on the repositories of the 21 projects with the PySZZ implementation [26]. The same list of bug-fixing commits will be given to all variants, and every commit of the dataset will receive a label from each variant. These labels will be compared with the reference labels. For each variant the precision and the recall will be computed, together with the rate at which clean commits are labelled as defect-introducing and the rate at which defect-introducing commits are labelled as clean. The agreement between the variants will be measured with Cohen’s κ [36], to see whether the variants make the same errors.

In the second study, prediction models will be trained on the reference labels and on the labels of each SZZ variant. Three models of different complexity are planned: LAPredict [18], a random forest model in the style of JITLine [19], and the online learner ORB [12]. The models will be evaluated in settings of increasing realism: random k -fold cross-validation, a chronological split, and prequential evaluation in which the labels are delivered with the delay reconstructed from the data, using the waiting period of 90 days proposed by Cabral et al. [12]. The predictions will be scored against the reference labels and, for comparison, against the training labels. The Matthews correlation coefficient will be used as the main performance measure, since it is suitable for imbalanced data [33, 34]. The comparisons will be made for each project and tested with paired non-parametric tests over the 21 projects.

5.2.2 Third stage: mechanism of the noise effect

The third stage addresses RQ3. If the results of the second stage show that SZZ labels reduce the performance of an online learner, the next question is which type of labelling error causes the loss. Two label sources differ in the number of false positives, in the number of false negatives and in the arrival times of the labels at the same time, so a comparison of label sources cannot answer this question. Experiments are therefore planned in which one property of the labels is changed at a time. Label noise will be injected into the reference labels at the rates measured in the second stage, and the labels of an SZZ variant will be corrected with the help of the reference labels, one type of error at a time. It will also be examined whether the oversampling used by online learners for imbalanced streams (Section 2.6) increases the effect of wrongly labelled positive examples.

5.2.3 Fourth stage: noise-aware online learner

The fourth stage addresses RQ4. Based on the findings of the third stage, a modification of the online learner will be designed that makes it less sensitive to the type of error found to be harmful. The modification has to work without access to clean labels, since these are not available in practice. It should improve the performance when the learner is trained on SZZ labels, and it should not reduce the performance when the labels are clean. The writing of the thesis will be completed in this stage.

5.3 Timeline

Table 5.1 gives the timeline of the thesis work.

Table 5.1: Timeline of the thesis work.

Stage	Work	Status
First	Literature survey; identification of research gaps; research questions and objectives; selection and preparation of the dataset	Completed
Second	Generation of SZZ labels; label quality analysis (RQ1); impact on prediction models under different evaluation settings (RQ2)	Planned
Third	Noise injection and label repair experiments; mechanism of the noise effect (RQ3)	Planned
Fourth	Design and evaluation of a noise-aware online learner (RQ4); thesis writing	Planned

References

- [1] T. Hall, S. Beecham, D. Bowes, D. Gray, and S. Counsell, “A systematic literature review on fault prediction performance in software engineering,” *IEEE Transactions on Software Engineering*, vol. 38, no. 6, pp. 1276–1304, 2012.
- [2] S. S. Rathore and S. Kumar, “A study on software fault prediction techniques,” *Artificial Intelligence Review*, vol. 51, no. 2, pp. 255–327, 2019.
- [3] Y. Kamei, E. Shihab, B. Adams, A. E. Hassan, A. Mockus, A. Sinha, and N. Ubayashi, “A large-scale empirical study of just-in-time quality assurance,” *IEEE Transactions on Software Engineering*, vol. 39, no. 6, pp. 757–773, 2013.
- [4] A. Mockus and D. M. Weiss, “Predicting risk of software changes,” *Bell Labs Technical Journal*, vol. 5, no. 2, pp. 169–180, 2000.
- [5] S. Kim, E. J. Whitehead, Jr., and Y. Zhang, “Classifying software changes: Clean or buggy?” *IEEE Transactions on Software Engineering*, vol. 34, no. 2, pp. 181–196, 2008.
- [6] J. Śliwerski, T. Zimmermann, and A. Zeller, “When do changes induce fixes?” in *Proceedings of the International Workshop on Mining Software Repositories (MSR)*, 2005, pp. 1–5.
- [7] S. Herbold, A. Trautsch, B. Ledel, *et al.*, “A fine-grained data set and analysis of tangling in bug fixing commits,” *Empirical Software Engineering*, vol. 27, no. 6, article 125, 2022.
- [8] D. A. da Costa, S. McIntosh, W. Shang, U. Kulesza, R. Coelho, and A. E. Hassan, “A framework for evaluating the results of the SZZ approach for identifying bug-introducing changes,” *IEEE Transactions on Software Engineering*, vol. 43, no. 7, pp. 641–657, 2017.
- [9] G. Rosa, L. Pascarella, S. Scalabrino, R. Tufano, G. Bavota, M. Lanza, and R. Oliveto, “Evaluating SZZ implementations through a developer-informed oracle,” in *Proceedings of the 43rd IEEE/ACM International Conference on Software Engineering (ICSE)*, 2021.
- [10] D. Falessi, J. Huang, L. Narayana, J. F. Thai, and B. Turhan, “On the need of preserving order of data when validating within-project defect classifiers,” *Empirical Software Engineering*, vol. 25, pp. 4805–4830, 2020.
- [11] M. Tan, L. Tan, S. Dara, and C. Mayeux, “Online defect prediction for imbalanced data,” in *Proceedings of the 37th International Conference on Software Engineering (ICSE)*, vol. 2, 2015, pp. 99–108.
- [12] G. G. Cabral, L. L. Minku, E. Shihab, and S. Mujahid, “Class imbalance evolution and verification latency in just-in-time software defect prediction,” in *Proceedings of the 41st International Conference on Software Engineering (ICSE)*, 2019, pp. 666–676.
- [13] G. G. Cabral and L. L. Minku, “Towards reliable online just-in-time software defect prediction,” *IEEE Transactions on Software Engineering*, vol. 49, no. 3, pp. 1342–1358, 2023.
- [14] Y. Zhao, K. Damevski, and H. Chen, “A systematic survey of just-in-time software defect prediction,” *ACM Computing Surveys*, vol. 55, no. 10, pp. 1–35, 2023.
- [15] S. McIntosh and Y. Kamei, “Are fix-inducing changes a moving target? A longitudinal case study of just-in-time defect prediction,” *IEEE Transactions on Software Engineering*, vol. 44, no. 5, pp. 412–428, 2018.
- [16] T. Hoang, H. K. Dam, Y. Kamei, D. Lo, and N. Ubayashi, “DeepJIT: An end-to-end deep learning framework for just-in-time defect prediction,” in *Proceedings of the 16th Interna-*

- tional Conference on Mining Software Repositories (MSR)*, 2019.
- [17] T. Hoang, H. J. Kang, D. Lo, and J. Lawall, “CC2Vec: Distributed representations of code changes,” in *Proceedings of the 42nd International Conference on Software Engineering (ICSE)*, 2020.
 - [18] Z. Zeng, Y. Zhang, H. Zhang, and L. Zhang, “Deep just-in-time defect prediction: How far are we?” in *Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2021.
 - [19] C. Pornprasit and C. Tantithamthavorn, “JITLine: A simpler, better, faster, finer-grained just-in-time defect prediction,” in *Proceedings of the 18th International Conference on Mining Software Repositories (MSR)*, 2021, pp. 369–379.
 - [20] Y. Yang, Y. Zhou, J. Liu, Y. Zhao, H. Lu, L. Xu, B. Xu, and H. Leung, “Effort-aware just-in-time defect prediction: Simple unsupervised models could be better than supervised models,” in *Proceedings of the 24th ACM SIGSOFT International Symposium on Foundations of Software Engineering (FSE)*, 2016, pp. 157–168.
 - [21] G. Rodríguez-Pérez, G. Robles, A. Serebrenik, A. Zaidman, D. M. Germán, and J. M. González-Barahona, “How bugs are born: A model to identify how bugs are introduced in software components,” *Empirical Software Engineering*, vol. 25, pp. 1294–1340, 2020.
 - [22] S. Kim, T. Zimmermann, K. Pan, and E. J. Whitehead, Jr., “Automatic identification of bug-introducing changes,” in *Proceedings of the 21st IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2006, pp. 81–90.
 - [23] S. Davies, M. Roper, and M. Wood, “Comparing text-based and dependence-based approaches for determining the origins of bugs,” *Journal of Software: Evolution and Process*, vol. 26, no. 1, pp. 107–139, 2014.
 - [24] E. C. Neto, D. A. da Costa, and U. Kulesza, “The impact of refactoring changes on the SZZ algorithm: An empirical study,” in *Proceedings of the 25th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, 2018, pp. 380–390.
 - [25] G. Rodríguez-Pérez, G. Robles, and J. M. González-Barahona, “Reproducibility and credibility in empirical software engineering: A case study based on a systematic literature review of the use of the SZZ algorithm,” *Information and Software Technology*, vol. 99, pp. 164–176, 2018.
 - [26] G. Rosa, L. Pascarella, S. Scalabrino, R. Tufano, G. Bavota, M. Lanza, and R. Oliveto, “A comprehensive evaluation of SZZ variants through a developer-informed oracle,” *Journal of Systems and Software*, vol. 202, article 111729, 2023.
 - [27] K. Herzig and A. Zeller, “The impact of tangled code changes,” in *Proceedings of the 10th Working Conference on Mining Software Repositories (MSR)*, 2013, pp. 121–130.
 - [28] S. Kim, H. Zhang, R. Wu, and L. Gong, “Dealing with noise in defect prediction,” in *Proceedings of the 33rd International Conference on Software Engineering (ICSE)*, 2011, pp. 481–490.
 - [29] C. Tantithamthavorn, S. McIntosh, A. E. Hassan, A. Ihara, and K. Matsumoto, “The impact of mislabelling on the performance and interpretation of defect prediction models,” in *Proceedings of the 37th International Conference on Software Engineering (ICSE)*, 2015, pp. 812–823.
 - [30] Y. Fan, X. Xia, D. A. da Costa, D. Lo, A. E. Hassan, and S. Li, “The impact of changes mislabeled by SZZ on just-in-time defect prediction,” *IEEE Transactions on Software Engineering*, vol. 47, no. 8, pp. 1559–1586, 2021.
 - [31] J. Gama, R. Sebastião, and P. P. Rodrigues, “On evaluating stream learning algorithms,” *Machine Learning*, vol. 90, no. 3, pp. 317–346, 2013.
 - [32] S. Tabassum, L. L. Minku, D. Feng, G. G. Cabral, and L. Song, “An investigation of cross-project learning in online just-in-time software defect prediction,” in *Proceedings of the 42nd International Conference on Software Engineering (ICSE)*, 2020, pp. 554–565.

- [33] J. Yao and M. Shepperd, "Assessing software defection prediction performance: Why using the Matthews correlation coefficient matters," in *Proceedings of the 24th International Conference on Evaluation and Assessment in Software Engineering (EASE)*, 2020, pp. 120–129.
- [34] B. W. Matthews, "Comparison of the predicted and observed secondary structure of T4 phage lysozyme," *Biochimica et Biophysica Acta (BBA) – Protein Structure*, vol. 405, no. 2, pp. 442–451, 1975.
- [35] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, article 6, 2020.
- [36] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.
- [37] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [38] N. C. Oza and S. Russell, "Online bagging and boosting," in *Proceedings of the 8th International Workshop on Artificial Intelligence and Statistics (AISTATS)*, 2001, pp. 229–236.
- [39] S. Wang, L. L. Minku, and X. Yao, "Resampling-based ensemble methods for online class imbalance learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 27, no. 5, pp. 1356–1368, 2015.
- [40] N. Natarajan, I. S. Dhillon, P. Ravikumar, and A. Tewari, "Learning with noisy labels," in *Advances in Neural Information Processing Systems (NIPS)*, 2013.
- [41] B. Han, Q. Yao, X. Yu, G. Niu, M. Xu, W. Hu, I. Tsang, and M. Sugiyama, "Co-teaching: Robust training of deep neural networks with extremely noisy labels," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [42] C. G. Northcutt, L. Jiang, and I. L. Chuang, "Confident learning: Estimating uncertainty in dataset labels," *Journal of Artificial Intelligence Research*, vol. 70, pp. 1373–1411, 2021.
- [43] H. Keshavarz and M. Nagappan, "ApacheJIT: A large dataset for just-in-time defect prediction," in *Proceedings of the 19th International Conference on Mining Software Repositories (MSR)*, 2022.
- [44] P. Mahbub, O. Shuvo, and M. M. Rahman, "Defectors: A large, diverse Python dataset for defect prediction," in *Proceedings of the 20th International Conference on Mining Software Repositories (MSR)*, 2023.
- [45] C. Ni, W. Wang, K. Yang, X. Xia, K. Liu, and D. Lo, "The best of both worlds: Integrating semantic features with expert features for defect prediction and localization," in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2022, pp. 672–683.